

Firdaus Soberi

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Education

- The University of Edinburgh, UK** 2022–2026 (Expected)
PhD in Particle Physics *ATLAS Experiment*
 - Thesis: *Search for CP Violation in the Vector-Boson-Fusion Production of the Higgs Boson in the $H \rightarrow \gamma\gamma$ Decay Channel with the ATLAS Detector*
 - Advisor (Co-Supervisor): *Prof. Philip Clark (Prof. Victoria Martin)*
- Osaka University, Japan** 2016–2018
MSc in Physics *CANDLES Experiment*
 - Thesis: *Measurement of β and γ Coincidence Spectrum of ^{24}Na for CANDLES Calibration System*
 - Advisor (Co-Supervisor): *Prof. Tadafumi Kishimoto (Prof. Sei Yoshida)*
- University of Washington Seattle, USA** 2012–2015
BSc (Hons) in Physics
 - Undergraduate Supervisor: *Prof. Shih-Chieh Hsu (ATLAS)*

Research Experience

- ATLAS Experiment – Higgs Physics** 2022–2026
The University of Edinburgh *UK*
 - Played a key role in the Run-3 $H \rightarrow \gamma\gamma$ analysis of CP violation in Vector-Boson Fusion, developing the analysis framework, machine-learning classification, event categorisation, and systematic uncertainty evaluation.
 - Contributed to Run-3 vertex reconstruction performance studies for $H \rightarrow \gamma\gamma$ under higher pile-up conditions.
 - Developed and validated muon punch-through simulation in ATLAS fast simulation, including a flow-matching generative approach.
- Research Assistant** March 2019–May 2020
Universiti Teknologi Malaysia *Malaysia*
 - Developed a statistical nuclear cascade model and curated data for proton and neutron emission following Ordinary Muon Capture, contributing to radioactive isotope production studies at the MuSIC experiment.
- Graduate Researcher – CANDLES** October 2016–September 2018
Graduate School of Science, Osaka University *Japan*
 - Contributed to detector calibration and position-alignment studies for the CANDLES neutrinoless double beta decay experiment.
- Research Assistant** December 2015–June 2016
Universiti Malaya *Malaysia*
 - Worked with Prof. Hassan Abu Kassim in the Theoretical Physics group on flavor-changing processes under the Malaysian Frontier Science Grant.
- CERN Winter Research Experience for Undergraduates** January 2015–April 2015
University of Michigan in partnership with CERN *CERN*
 - Worked on ATLAS trigger and track-based missing transverse energy studies in $Z(\nu\nu)h(b\bar{b})$, and contributed to an anomalous quartic gauge coupling analysis.
- Undergraduate Researcher** June 2014–August 2015
University of Washington *USA*
 - Studied BSM phenomenology and ATLAS physics using MadGraph aMC@NLO, including 2HDM, anomalous quartic gauge couplings, and Monte Carlo studies of the ATLAS insertable B-layer.

Other Experience

Full Stack Developer

June 2019–May 2022

CMA Solutions Sdn. Bhd.

Malaysia

- Designed and developed cross-platform mobile applications and full-stack software systems for warehouse and logistics operations, including backend APIs and database systems.
- Developed software solutions for logistics workflow automation and operational planning.

Publications

- ATLAS Collaboration. (2026). Search for anomalies in vector-boson fusion production of the Higgs boson in $H(\rightarrow \gamma\gamma)jj$ events using 164 fb^{-1} of pp collision data collected at $\sqrt{s} = 13.6\text{ TeV}$ with the ATLAS detector. [arXiv:2603.20087](#) (Accepted for publication).
- F. Soberi, C. Jiang, P. Clark and T. Qiu. (2026). “Generative Muon Punch-Through with Flow-Matching in ATLAS”. *In preparation for CHEP 2026 proceedings*. [CHEP 2026 abstract](#).
- A. Bohare, F. Soberi and C. Jiang. (2026). “BitNet-based Quantized Neural Networks for Efficient Inference in High-Energy Physics”, *DAE Symp. Nucl. Phys.* **69**, 1229–1230.
- T. Novak *et al.*, ATLAS computing activity. (2026). “The ATLAS simulation R&D programme for Run 4”, *PoS LHCP2025*, 126.
- F. A. G. Corchia *et al.*, ATLAS computing activity. (2025). “AtlFast3: Fast Simulation in ATLAS for LHC Run 3 and Beyond”, *EPJ Web Conf.* **337**, 01355.
- F. Soberi, ATLAS Collaboration. (2025). “Higgs boson mass, width and CP measurements with the ATLAS detector”, *PoS DIS2025* (2025), 059.
- F. Othman, I. H. Hashim, H. Ejiri, R. Razali, F. Ibrahim and F. Soberi. (2021). “Proton neutron emission model for Muon charge exchange reaction”, *AIP Conf. Proc.* **2319** no.1, 080004.
- I. H. Hashim, H. Ejiri, F. Othman, F. Ibrahim, F. Soberi, N. N. A. M. A. Ghani, T. Shima, A. Sato and K. Ninomiya. (2020). “Nuclear Isotope Production by Ordinary Muon Capture Reaction”, *Nucl. Instrum. Meth. A*, 163749.
- S. Umehara *et al.*. (2020). “Search for neutrino-less double beta decay of ^{48}Ca -CANDLES-”, *J. Phys. Conf. Ser.* **1643** no.1, 012028.
- K. Tetsuno *et al.*, CANDLES. (2020). “Status of ^{48}Ca double beta decay search and its future prospect in CANDLES”, *J. Phys. Conf. Ser.* **1468** no.1, 012132.

Research Presentations

- [CHEP 2026](#), Chulalongkorn University, Bangkok, Thailand. (25–29 May 2026), “Generative Muon Punch-Through with Flow-Matching in ATLAS” (*Talk*).
- [ATLAS Software & Computing Week #83](#), CERN. (9–13 February 2026), “FastCaloSim Group Progress Report” (*Invited Talk*).
- [ATLAS UK Annual Meeting](#), Sussex. (7–9 January 2026), “Software & Computing Activities” on behalf of UK colleagues (*Invited Talk*).
- [Institute of Physics \(IOP\) Joint APP and HEPP Annual Conference 2025](#), University of Cambridge. (7–9 April 2025), “Test of CP-invariance of the Higgs boson in vector-boson fusion production in the $H \rightarrow \gamma\gamma$ channel with the ATLAS detector” (*Talk*).
- [Deep Inelastic Scattering \(DIS\) 2025](#), Cape Town, South Africa. (24–28 March 2025), “Measurements of Higgs boson properties (mass, width, and Spin/CP) with the ATLAS detector” on behalf of the ATLAS collaboration (*Invited Talk*).

- ATLAS Collaboration Week, CERN. (17–21 February 2025), “Beyond ISF: Muon Punch Through Simulation in AtlFast3” (*Lightning Talk*).
- ATLAS UK Annual Meeting, UCL. (6–8 January 2025), “Higgs to diphoton” analyses on behalf of UK colleagues (*Invited Talk*).

Teaching Experience

- Thermal Physics (PHYS09061). TA – Semester 2, 2025/2026
- Linear Algebra and Several Variable Calculus (PHYS08042). TA – Semester 1, 2025/2026
- Programming and Data Analysis: Python (PHYS08049). TA – Semester 1, 2025/2026
- Data Analysis and Machine Learning (PGPH11105). TA – Semester 2, 2024/2025
- Experimental Physics 2 (PHYS08058). TA – Semester 2, 2023/2024
- Physics 1B: The Stuff of the Universe (PHYS08017). TA – Semester 2, 2023/2024
- Relativity, Nuclear and Particle Physics (PHYS10096). TA – Semester 1, 2023/2024
- Physics 1A: Foundations (PHYS08016). TA – Semester 1, 2023/2024

Awards & Scholarships

- Graduate Excellence Programme, MARA Agency, Malaysia (2022–2026).
- Sumitomo Foundation Scholarship, Japan (2017–2018).
- Mary Gates Research Scholarship, University of Washington (2015).
- Malaysian Public Service Department Scholarship, Malaysia (2012–2015).

Schools

- Asia-Europe-Pacific School of High-Energy Physics (AEP SHEP 2024), Nakhon Pathom, Thailand. (12–25 June 2024), “Muon Punch-through Simulation in ATLAS” (*Poster*)
- STFC HEP School 2023, University of Oxford. (3–15 September 2023), “Vertex Finding for $H \rightarrow \gamma\gamma$ in Run-3 Using Neural Networks” (*Poster*)

Detector Operations

- Completed 376 hours of ATLAS detector operations shifts during a year-long attachment at CERN, including responsibilities in Run Control, Trigger, and Data Quality.